

Katie Olsen

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EDUCATION

Bachelor of Science, Computer Engineering

Brigham Young University • Provo, UT • April 2026 • 3.88

EXPERIENCE

Engineering Intern

Browns Hill Engineering and Controls

May 2024 – August 2024, Littleton, CO

- Consolidated multiple internal certificate-signing platforms into a single Ignition-based application, streamlining workflows (CSR creation, signing, revocation, storage, and certificate tree management) and reducing tool fragmentation across teams.
- Developed a Python SDK to interface with Cloudflare's SSL certificate APIs, using Postman for endpoint validation and integration testing, and enabling secure certificate operations within an Ignition-based user interface.
- Designed and implemented an internal certificate management tool in Ignition, including UI wireframes and documentation, and transitioned the system to an engineering stakeholder for deployment and ongoing use.

Engineering Intern

Browns Hill Engineering and Controls

May 2023 – August 2023, Littleton, CO

- Developed an internal 'data vault' application in Ignition and Python to consolidate duplicated IT, field, and engineering data from disparate Excel-based sources into a centralized MySQL-backed system, improving data consistency and accessibility across teams.
- Built database integration pipelines in Python to map MySQL data into structured models and classes, enabling real-time display and interaction through an Ignition-based user interface.
- Refactored and modernized industrial control interfaces from FactoryTalk to Ignition, updating PLC tag connections and backend logic while preserving system functionality and improving usability.

Engineering Intern

Browns Hill Engineering and Controls

May 2021 – September 2021, Littleton, CO

- Developed Ignition Python scripting tools to convert industrial PLC/HMI tag data from multiple systems (e.g. FactoryTalk, Studio5000) from CSV formats into structured JSON, reducing manual data processing time from hours/days to automated workflows.
- Researched and documented internal engineering tooling and best practices using Linux and virtual machines, including certificate-signing platforms, SCADA systems, and industrial radio debugging procedures.
- Presented documentation on cybersecurity practices and engineering workflows to 10-15 engineers in recurring technical briefings.

INVOLVEMENT

Automated Pit Exploration System Capstone Project

BYU Capstone Team Project for Idaho National Laboratory (INL) • Programmer

September 2025 – April 2026

- Developed a Unity-based robotic control interface in C#, designing operator-focused UI components and real-time robot visualizations based on feedback from field users and project stakeholders.
- Integrated ROS2 communication between Unity and physical robots from Idaho National Laboratory and Florida International University using WSL, ROS#, and rosbriidge to enable real-time control and bidirectional communication.
- Managed project documentation and sponsor handoff materials, creating organized GitHub READMEs and technical documentation to support long-term maintainability and usability.

Brigham Young University – Antweight Battle Bot Competition

Co-team leader • Programmer

December 2023

- Programmed an ESP32 in Arduino and designed the motor-control electronics for an Antweight competition-ready Battle Bot.
- Led a team of 4 by organizing team meetings, delegating tasks, and coordinating project development to ensure competition readiness.

The Church of Jesus Christ of Latter-Day Saints

Full Time Missionary • Teaching and Leadership Experience

September 2021 – March 2023

- Delivered daily lessons and presentations in small-group and congregation settings, speaking to audiences ranging from 2-100+ individuals.
- Collaborated with community and congregational leaders to identify needs and coordinate support efforts for individuals and families
- Taught weekly English classes for Spanish speakers and participated in biweekly service activities including food banks and park-cleanup.

Lockheed Martin Code Quest Competition

Programmer

April 2019

- Solved 20-30 timed coding challenges of varying difficulty in a team of 3 using Java.

SKILLS

Programming Languages: C, C++, C#, Python, Java, JavaScript, TypeScript, RISC-V Assembly, SystemVerilog

Web and Cloud Technologies: HTML, CSS, AWS, Postman

Databases: MySQL, NoSQL

Systems and DevOps: Linux, Docker, Virtual Machines, Git/GitHub

Embedded and Hardware: Arduino, ESP32, FPGA, PYNQ, circuit board design, ROS2

Industrial and Engineering Tools: Ignition (HMI/SCADA), FactoryTalk, Unity Game Engine, MATLAB

Languages: English (Native), Spanish (Advanced Proficiency)